Vectors: Components and Operations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Vector arithmetic

Let $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 4 \rangle$. Find:

a $\mathbf{u} + \mathbf{v} = \langle 2, 2 \rangle$ **b** $3\mathbf{u} - 2\mathbf{v} = \langle 11, -14 \rangle$ **c** $|\mathbf{u}| = \sqrt{13}$

(a) $= \langle 2, 2 \rangle$; (b) $= \langle 11, -14 \rangle$; (c) $= \sqrt{13}$

2 [Foundational] • Target: Vector arithmetic

Find the component form and magnitude of the vector from $P(1, 3)$ to $Q(5, -2)$.

$\vec{PQ} = \langle 4, -5 \rangle$, $|\vec{PQ}| = \sqrt{16 + 25} = \sqrt{41}$.

3 [Foundational] • Target: Applications

A plane flies due east at 200 mph while the wind blows due north at 50 mph.

a Find the magnitude of the plane's resultant velocity, to one decimal place.
$\sqrt{200^2 + 50^2} = \sqrt{42500} \approx 206.2$ mph.

b Find the direction of the resultant, as an angle north of east to two decimal places.
$\theta = \tan^{-1}\left(\frac{50}{200}\right) \approx 14.04°$ north of east.

(a) $\sqrt{200^2 + 50^2} = \sqrt{42500} \approx 206.2$ mph.; (b) $\theta = \tan^{-1}\left(\frac{50}{200}\right) \approx 14.04°$ north of east.

4 [Foundational] • Target: Applications

A boat aims to cross a river by heading due north at 4 m/s, but the current pushes it east at 1.5 m/s.

a Find the boat's resultant speed, to two decimal places. $\sqrt{4^2 + 1.5^2} = \sqrt{18.25} \approx 4.27$ m/s.

b If the river is 200 m wide, find how long the crossing takes. Only the northward component moves it across: $t = \frac{200}{4} = 50$ s.

c Find how far downstream (east) the boat lands. Drift $= 1.5 \times 50 = 75$ m.

(a) $\sqrt{4^2 + 1.5^2} = \sqrt{18.25} \approx 4.27$ m/s.; (b) Only the northward component moves it across: $t = \frac{200}{4} = 50$ s.; (c) Drift $= 1.5 \times 50 = 75$ m.

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5 [Foundational] • Target: Position vectors and geometry

Given $A(-2, 5)$ and $B(4, -3)$, find the midpoint of $\overline{AB}$ and the vector $\vec{AB}$.

Midpoint: $\left(\frac{-2+4}{2}, \frac{5-3}{2}\right) = (1, 1)$. $\vec{AB} = \langle 6, -8 \rangle$.

6 [Foundational] • Target: More applications

Given $\mathbf{u} = \langle 4, -1 \rangle$, find a vector orthogonal to $\mathbf{u}$ with the same magnitude as $\mathbf{u}$.

Swap components and negate one: $\langle 1, 4 \rangle$. Check: $\mathbf{u} \cdot \langle 1, 4 \rangle = 4 - 4 = 0$ (orthogonal), and $|\langle 1, 4 \rangle| = \sqrt{17} = |\mathbf{u}|$.

7 [Foundational] • Target: Finding direction angles

Find the direction angle of $\mathbf{v} = \langle -3, 3\sqrt{3} \rangle$, to the nearest degree.

$\theta = \tan^{-1}\left(\frac{3\sqrt{3}}{-3}\right)$ with reference angle $60°$, but since the vector has negative x and positive y (Quadrant II): $\theta = 180° - 60° = 120°$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Parallel and equal vectors

Determine whether $\mathbf{u} = \langle 4, -6 \rangle$ and $\mathbf{v} = \langle -6, 9 \rangle$ are parallel, and if so, state whether they point in the same or opposite directions.

$\mathbf{v} = -\frac{3}{2}\mathbf{u}$ (check: $-\frac{3}{2}(4) = -6$ ✓, $-\frac{3}{2}(-6) = 9$ ✓). Parallel, with a negative scalar multiple, so they point in OPPOSITE directions.

9 [Intermediate] • Target: Vectors in three real-world scenarios

A hiker walks 3 km east, then 4 km north, then 2 km west. Find the magnitude of the hiker's total displacement from the starting point.

Net displacement: $\langle 3 - 2, 4 \rangle = \langle 1, 4 \rangle$. Magnitude $= \sqrt{1 + 16} = \sqrt{17} \approx 4.12$ km.

10 [Intermediate] • Target: Vectors in three real-world scenarios

Three ropes pull on a ring with forces $\langle 10, 0 \rangle$, $\langle -5, 8.66 \rangle$, and $\langle -5, -8.66 \rangle$ N. Find the net force, and explain what this implies physically.

Sum: $\langle 10 - 5 - 5, 0 + 8.66 - 8.66 \rangle = \langle 0, 0 \rangle$. The net force is zero, meaning the ring is in equilibrium (the three forces are $120°$ apart and balance exactly).

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11 [Intermediate] • Target: Vectors in three real-world scenarios

A crane hoists a load using two cables. One cable exerts a force of $\langle -200, 350\rangle$ N and the other exerts $\langle 200, 400\rangle$ N. Find the combined lifting force (the vertical component of the resultant), and explain why the horizontal components matter for the crane's structural design even though they cancel.

Resultant: $\langle -200+200,\ 350+400\rangle = \langle 0, 750\rangle$. The vertical (lifting) force is 750 N. Although the horizontal components cancel in the resultant, each individual cable still experiences its own horizontal tension, which the crane's frame must be strong enough to withstand even though the net horizontal effect on the load is zero.

12 [Intermediate] • Target: Vectors in three real-world scenarios

A swimmer aims to swim directly across a river at 1.5 m/s relative to the water, while the current flows at 0.8 m/s downstream. Find the swimmer's resultant speed relative to the shore, and the angle (from the intended straight-across direction) at which they actually drift, to the nearest degree.

Resultant speed $= \sqrt{1.5^2 + 0.8^2} = \sqrt{2.25 + 0.64} = \sqrt{2.89} = 1.7$ m/s. Drift angle $= \tan^{-1}\left(\frac{0.8}{1.5}\right) \approx 28°$ from the intended direction.

13 [Intermediate] • Target: Vector arithmetic

Find the unit vector in the direction of $\mathbf{v} = \langle 6,\ -8\rangle$.

$|\mathbf{v}| = 10$, so $\hat{\mathbf{v}} = \left\langle \frac{3}{5},\ -\frac{4}{5}\right\rangle$.

14 [Intermediate] • Target: Vector arithmetic

Write the vector of magnitude 12 making an angle of $150°$ with the positive x-axis in component form (exact values).

$\langle 12\cos 150°, 12\sin 150°\rangle = \langle -6\sqrt{3}, 6\rangle$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Applications • Cross-Lesson: 1.01 Functions and Function Notation

Two forces $\mathbf{F}_1 = \langle 30, 0\rangle$ N and $\mathbf{F}_2 = \langle 0, 40\rangle$ N act on an object. Find the magnitude of the resultant force.

$|\mathbf{F}_1 + \mathbf{F}_2| = |\langle 30, 40\rangle| = 50$ N.

16 [Challenging] • Target: Position vectors and geometry • Cross-Lesson: 1.02 Domain and Range

Let $\mathbf{a} = \langle 2, 3\rangle$ and $\mathbf{b} = \langle -4, 1\rangle$. Find a vector $\mathbf{c}$ such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \langle 0, 0\rangle$, and verify your answer.

$\mathbf{c} = -(\mathbf{a} + \mathbf{b}) = -\langle -2, 4\rangle = \langle 2,\ -4\rangle$. Check: $\langle 2, 3\rangle + \langle -4, 1\rangle + \langle 2,\ -4\rangle = \langle 0, 0\rangle$. ✓

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17 [Challenging] • Target: Position vectors and geometry • Cross-Lesson: 1.03 Average Rate of Change

Express $\mathbf{v} = \langle 5, -12 \rangle$ in terms of the standard unit vectors $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{v}|$.

$\mathbf{v} = 5\mathbf{i} - 12\mathbf{j}$. $|\mathbf{v}| = \sqrt{25 + 144} = 13$.

18 [Challenging] • Target: More applications • Cross-Lesson: 2.01 Right-Triangle Trigonometry

An airplane has an airspeed of 305 mph and flies on a heading $7.5°$ west of due north, while a 40 mph wind blows due east. Find the plane's resultant ground velocity vector, and its ground speed, to one decimal place (take east as the positive x-direction and north as the positive y-direction).

Plane's own velocity: $\langle -305\sin 7.5°,\ 305\cos 7.5° \rangle \approx \langle -39.8,\ 302.4 \rangle$. Adding the wind $\langle 40, 0 \rangle$: resultant $\approx \langle 0.2,\ 302.4 \rangle$. Ground speed $\approx \sqrt{0.2^2 + 302.4^2} \approx 302.4$ mph — very nearly due north, as intended.

19 [Challenging] • Target: Finding direction angles • Cross-Lesson: 2.02 Radian Measure and Arc Length

A vector has magnitude 10 and direction angle $200°$. Write it in component form, rounded to two decimal places.

$\langle 10\cos 200°, 10\sin 200° \rangle \approx \langle -9.40,\ -3.42 \rangle$.

20 [Challenging] • Target: Vector equations and unknowns • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Find scalars c_1 and c_2 such that $c_1\langle 1, 0 \rangle + c_2\langle 0, 1 \rangle = \langle -3, 7 \rangle$.

Since $\langle 1, 0 \rangle$ and $\langle 0, 1 \rangle$ are the standard basis vectors, $c_1 = -3$ and $c_2 = 7$ directly.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Geometric vs Component Form

Find the component form of vector $\mathbf{v}$ with initial point $P(2, -3)$ and terminal point $Q(-1, 4)$.

a. $\langle -3, 7 \rangle$ ✓

b. $\langle 1, 1 \rangle$

c. $\langle 3, -7 \rangle$

d. $\langle -2, -12 \rangle$

$\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle = \langle -1 - 2, 4 - (-3) \rangle = \langle -3, 7 \rangle$.

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22 [Foundational] • Target: Geometric vs Component Form

Express $\mathbf{v} = \langle 5, -8 \rangle$ in terms of the standard unit basis vectors $\mathbf{i}$ and $\mathbf{j}$.

a. $5\mathbf{i} - 8\mathbf{j}$ ✓

b. $5\mathbf{i} + 8\mathbf{j}$

c. $-8\mathbf{i} + 5\mathbf{j}$

d. $5\mathbf{j} - 8\mathbf{i}$

$\mathbf{i} = \langle 1, 0 \rangle$ and $\mathbf{j} = \langle 0, 1 \rangle$, so $\langle 5, -8 \rangle = 5\mathbf{i} - 8\mathbf{j}$.

23 [Foundational] • Target: Geometric vs Component Form

When are two geometric vectors considered equal?

a. **When they have both the same magnitude and the same direction** ✓

b. When they have the same initial point

c. When their lengths are equal regardless of direction

d. When they are collinear

Vectors are defined strictly by magnitude and direction, independent of their initial position in space.

24 [Foundational] • Target: Magnitude and Direction Angle

Find the magnitude of $\mathbf{v} = \langle -6, 8 \rangle$.

a. 10 ✓

b. 14

c. $\sqrt{28}$

d. 2

$||\mathbf{v}|| = \sqrt{(-6)^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.

25 [Foundational] • Target: Magnitude and Direction Angle

Find the direction angle $\theta \in [0^\circ, 360^\circ)$ of $\mathbf{v} = \langle -3, 3\sqrt{3} \rangle$.

a. 120° ✓

b. 60°

c. 150°

d. 240°

In Quadrant II: $\tan\theta = \frac{3\sqrt{3}}{-3} = -\sqrt{3}$. Reference angle is 60°, so $\theta = 180^\circ - 60^\circ = 120^\circ$.

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26 [Foundational] • Target: Magnitude and Direction Angle

Find the components of a vector with magnitude 12 and direction angle 210°.

a. $\langle -6\sqrt{3}, -6\rangle$ ✓

b. $\langle -6, -6\sqrt{3}\rangle$

c. $\langle 6\sqrt{3}, 6\rangle$

d. $\langle -6\sqrt{2}, -6\sqrt{2}\rangle$

$v_x = 12\cos(210^\circ) = 12(-\sqrt{3}/2) = -6\sqrt{3}$. $v_y = 12\sin(210^\circ) = 12(-1/2) = -6$.

27 [Foundational] • Target: Vector Addition, Subtraction, and Scalar Multiplication

Given $\mathbf{u} = \langle 3, -2\rangle$ and $\mathbf{v} = \langle -1, 5\rangle$, find $2\mathbf{u} - 3\mathbf{v}$.

a. $\langle 9, -19\rangle$ ✓

b. $\langle 3, 11\rangle$

c. $\langle 9, 11\rangle$

d. $\langle 3, -19\rangle$

$2\mathbf{u} = \langle 6, -4\rangle$. $3\mathbf{v} = \langle -3, 15\rangle$. Difference $= \langle 6-(-3), -4-15\rangle = \langle 9, -19\rangle$.

28 [Foundational] • Target: Vector Addition, Subtraction, and Scalar Multiplication

Find the unit vector $\mathbf{u}$ in the same direction as $\mathbf{v} = \langle 5, -12\rangle$.

a. $\langle 5/13, -12/13\rangle$ ✓

b. $\langle 5/17, -12/17\rangle$

c. $\langle 1, -1\rangle$

d. $\langle 5, -12\rangle$

$||\mathbf{v}|| = \sqrt{25+144} = 13$. Unit vector $\mathbf{u} = \frac{\mathbf{v}}{||\mathbf{v}||} = \langle 5/13, -12/13\rangle$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Vector Addition, Subtraction, and Scalar Multiplication

What is the result of multiplying any vector $\mathbf{v}$ by scalar -1?

a. A vector of the same magnitude pointing in the exactly opposite direction ✓

b. The zero vector

c. A scalar number

d. A perpendicular vector

Scalar multiplication by -1 negates every component, reversing the direction while preserving the magnitude.

30 [Intermediate] • Target: Resultant Forces and Navigation

Two forces act on an object: $\mathbf{F}_1 = \langle 20, 0 \rangle$ N and $\mathbf{F}_2 = \langle 0, 15 \rangle$ N. Find the magnitude of the resultant force.

a. 25 N ✓

b. 35 N

c. 5 N

d. 300 N

Resultant $\mathbf{F} = \langle 20, 15 \rangle$. Magnitude $||\mathbf{F}|| = \sqrt{20^2 + 15^2} = \sqrt{400 + 225} = \sqrt{625} = 25$ N.

31 [Intermediate] • Target: Resultant Forces and Navigation

An airplane flies at a heading due east with airspeed 400 mph. A wind blows from the south at 50 mph. Find the ground speed of the plane.

a. $\sqrt{400^2 + 50^2} \approx 403.1$ mph ✓

b. 450 mph

c. 350 mph

d. 425 mph

Plane velocity $\mathbf{v}_p = \langle 400, 0 \rangle$. Wind velocity (blowing from south means heading north) $\mathbf{v}_w = \langle 0, 50 \rangle$. Total $||\mathbf{v}|| = \sqrt{160000 + 2500} = \sqrt{162500} \approx 403.1$ mph.

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32 [Intermediate] • Target: Resultant Forces and Navigation

In the airplane problem, what is the drift angle north of east?

a. $\arctan(50/400) \approx 7.1^\circ$ ✓

b. 12.5°

c. 45°

d. 3.6°

$\tan\theta = \frac{50}{400} = 0.125 \Rightarrow \theta = \arctan(0.125) \approx 7.1^\circ$.

33 [Intermediate] • Target: Vector arithmetic

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?

a. $\langle 3, 4 \rangle$ ✓

b. $\langle -3, -4 \rangle$

c. $\langle 7, 10 \rangle$

d. $\langle 1, 4 \rangle$

To find the component form of a vector $\vec{PQ}$ from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$, we subtract the coordinates of the initial point P from the coordinates of the terminal point Q. The formula is $\vec{PQ} = \langle x_2 - x_1, y_2 - y_1 \rangle$. Given $P = (2, 3)$ and $Q = (5, 7)$, we have $x_1 = 2$, $y_1 = 3$, $x_2 = 5$, and $y_2 = 7$. Applying the formula: $\vec{PQ} = \langle 5 - 2, 7 - 3 \rangle$ $\vec{PQ} = \langle 3, 4 \rangle$ Therefore, the component form of the vector $\vec{PQ}$ is.

34 [Intermediate] • Target: Applications

Find the unit vector in the direction of $\mathbf{v} = \langle -8, 15 \rangle$.

a. $\left\langle \frac{9}{17}, \frac{15}{17} \right\rangle$

b. $\left\langle \frac{15}{17}, -\frac{8}{17} \right\rangle$

c. $\left\langle -\frac{8}{17}, \frac{15}{17} \right\rangle$ ✓

d. $\langle -8, 15 \rangle$

$|\mathbf{v}| = 17$. Unit vector $= \frac{\mathbf{v}}{|\mathbf{v}|} = \left\langle \frac{-8}{17}, \frac{15}{17} \right\rangle$.

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35 [Intermediate] • Target: Position vectors and geometry

Find the unit vector in the direction of $\mathbf{v} = \langle -6, 8 \rangle$.

a. $\left\langle \frac{2}{5}, \frac{4}{5} \right\rangle$

b. $\langle -6, 8 \rangle$

c. $\left\langle -\frac{3}{5}, \frac{4}{5} \right\rangle$ ✓

d. $\left\langle \frac{4}{5}, -\frac{3}{5} \right\rangle$

$|\mathbf{v}| = 10$. Unit vector $= \frac{\mathbf{v}}{|\mathbf{v}|} = \left\langle \frac{-6}{10}, \frac{8}{10} \right\rangle$.

36 [Intermediate] • Target: More applications

Find the unit vector in the direction of $\mathbf{v} = \langle 3, -4 \rangle$.

a. $\left\langle \frac{8}{5}, -\frac{4}{5} \right\rangle$

b. $\langle 3, -4 \rangle$

c. $\left\langle -\frac{4}{5}, \frac{3}{5} \right\rangle$

d. $\left\langle \frac{3}{5}, -\frac{4}{5} \right\rangle$ ✓

$|\mathbf{v}| = 5$. Unit vector $= \frac{\mathbf{v}}{|\mathbf{v}|} = \left\langle \frac{3}{5}, \frac{-4}{5} \right\rangle$.

37 [Intermediate] • Target: Finding direction angles

Consider two vectors, $\mathbf{u} = (3; k)$ and $\mathbf{v} = (-2; 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter k.

a. **-6** ✓

b. 6

c. 3

d. -3

Two vectors are parallel if one is a scalar multiple of the other. Therefore, there must exist a scalar c such that $\mathbf{u} = c\mathbf{v}$. This gives us the equations: $3 = c(-2)$ $k = c(4)$ From the first equation, we can solve for c: $c = \frac{3}{-2} = -\frac{3}{2}$ Now substitute this value of c into the second equation to find k: $k = \left(-\frac{3}{2}\right)(4)$ $k = -\frac{12}{2}$ $k = -6$ Therefore, the value of k is -6. A common algebraic trap is to mistakenly set the components equal with.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Vector equations and unknowns • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Consider the vector $\mathbf{v} = \langle 3, 4 \rangle$. What is the magnitude of this vector?

a. 5 ✓

b. 7

c. 12

d. 25

The magnitude of a vector $\mathbf{v} = \langle x, y \rangle$ is given by the formula $|\mathbf{v}| = \sqrt{x^2 + y^2}$. For the given vector $\mathbf{v} = \langle 3, 4 \rangle$, we have x = 3 and y = 4. Substituting these values into the formula: $|\mathbf{v}| = \sqrt{3^2 + 4^2}$ $|\mathbf{v}| = \sqrt{9 + 16}$ $|\mathbf{v}| = \sqrt{25}$ $|\mathbf{v}| = 5$ Therefore, the magnitude of the vector $\mathbf{v}$ is 5.

39 [Challenging] • Target: Parallel and equal vectors • Cross-Lesson: 2.02 Radian Measure and Arc Length

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. What are the components of the vector $\mathbf{v}$?

a. ■3, 4■ ✓

b. ■7, 10■

c. ■-3, -4■

d. ■4, 3■

The components of a vector from point $P = (x_1, y_1)$ to point $Q = (x_2, y_2)$ are given by $\langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, $P = (2, 3)$ and $Q = (5, 7)$. Therefore, the components of vector $\mathbf{v}$ are $\langle 5 - 2, 7 - 3 \rangle = \langle 3, 4 \rangle$.

40 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Find the angle between $\mathbf{u} = \langle 1, 3 \rangle$ and $\mathbf{v} = \langle 2, -2 \rangle$ (to one decimal place).

a. 116.6° ✓

b. 106.6°

c. 126.6°

d. 63.4°

$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{-4}{3.16 \times 2.83} \approx -0.447$, so $\theta \approx 116.6°$.

41 [Challenging] • Target: Geometric vs Component Form • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the angle between $\mathbf{u} = \langle 4, 1 \rangle$ and $\mathbf{v} = \langle 3, -1 \rangle$ (to one decimal place).

a. 147.5°

b. 42.5°

c. 22.5°

d. 32.5° ✓

$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{11}{4.12 \times 3.16} \approx 0.844$, so $\theta \approx 32.5°$.

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42 [Challenging] • Target: Magnitude and Direction Angle • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Find the angle between $\mathbf{u} = \langle 3, 2 \rangle$ and $\mathbf{v} = \langle 2, 2 \rangle$ (to one decimal place).

a. 1.3°

b. 21.3°

c. **11.3°** ✓

d. 168.7°

$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{10}{3.61 \times 2.83} \approx 0.981$, so $\theta \approx 11.3°$.

43 [Challenging] • Target: Vector Addition, Subtraction, and Scalar Multiplication • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the angle between $\mathbf{u} = \langle 2, 3 \rangle$ and $\mathbf{v} = \langle 2, 2 \rangle$ (to one decimal place).

a. 1.3°

b. 21.3°

c. 168.7°

d. **11.3°** ✓

$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{10}{3.61 \times 2.83} \approx 0.981$, so $\theta \approx 11.3°$.

44 [Challenging] • Target: Resultant Forces and Navigation • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the angle between $\mathbf{u} = \langle 3, 3 \rangle$ and $\mathbf{v} = \langle 3, 2 \rangle$ (to one decimal place).

a. 168.7°

b. 21.3°

c. 1.3°

d. **11.3°** ✓

$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{15}{4.24 \times 3.61} \approx 0.981$, so $\theta \approx 11.3°$.

45 [Challenging] • Target: Vector arithmetic • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Find the angle between $\mathbf{u} = \langle 3, 3 \rangle$ and $\mathbf{v} = \langle 4, 1 \rangle$ (to one decimal place).

a. **31.0°** ✓

b. 41.0°

c. 21.0°

d. 149.0°

$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{15}{4.24 \times 4.12} \approx 0.857$, so $\theta \approx 31.0°$.

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